

Chapter 7

Two-Sample Hypothesis Testing

Chapter 6 tested a single sample against a fixed claim: is our mean equal to some stated value μ_0 ? This chapter asks a more common question in practice: *do two groups differ from each other?*

Does a new teaching method improve test scores compared to the old one? Is the average salary higher in one city than another? Do two political candidates draw different levels of support? These are two-sample questions, and they appear constantly in research, medicine, business, and public policy.

The good news: the logic is identical to Chapter 6. We still compute a z -score, still find a p -value, still compare to α . The only change is the structure of the test statistic, which now measures the distance between *two* sample statistics rather than the distance of one statistic from a fixed claim.

7.1 The Setting: Two Independent Samples

Definition: Independent Samples. Two samples are **independent** if the selection of individuals in one sample has no influence on the selection in the other. The two groups are separate.

Examples of independent samples:

- Students randomly assigned to two different teaching methods.
- Voters in two different cities surveyed separately.
- Patients in a treatment group vs. a separate control group.

Definition: Paired (Dependent) Samples. Two samples are **paired** if each observation in one sample corresponds to a specific observation in the other.

Examples of paired samples:

- Pre-test and post-test scores for the same students.

- Blood pressure measured on the same patient before and after medication.
- Left-shoe and right-shoe measurements on the same people.

Common Pitfall. Always identify whether samples are independent or paired before choosing a test. Using an independent-samples test on paired data throws away information and reduces the power of the test. When in doubt, ask: *does each observation in Group 1 correspond naturally to a specific observation in Group 2?* If yes, the data are paired.

This chapter covers independent samples. The notation for the two groups uses subscripts 1 and 2:

	Sample size	Sample mean	Pop. mean	Sample SD	Pop. SD
Group 1	n_1	$\bar{x}_1$	μ_1	s_1	σ_1
Group 2	n_2	$\bar{x}_2$	μ_2	s_2	σ_2

7.2 The Sampling Distribution of $\bar{x}_1 - \bar{x}_2$

The key statistic is the **difference in sample means**: $\bar{x}_1 - \bar{x}_2$. If the two populations have the same mean ($\mu_1 = \mu_2$), then on average we expect $\bar{x}_1 - \bar{x}_2 = 0$. Observed differences far from zero are evidence that $\mu_1 \neq \mu_2$.

By the CLT applied to both samples simultaneously:

Key Formula. If both samples are large ($n_1 \geq 30$ and $n_2 \geq 30$) and independent:

$$\bar{x}_1 - \bar{x}_2 \sim N\left(\mu_1 - \mu_2, \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}\right).$$

The standard error of the difference is:

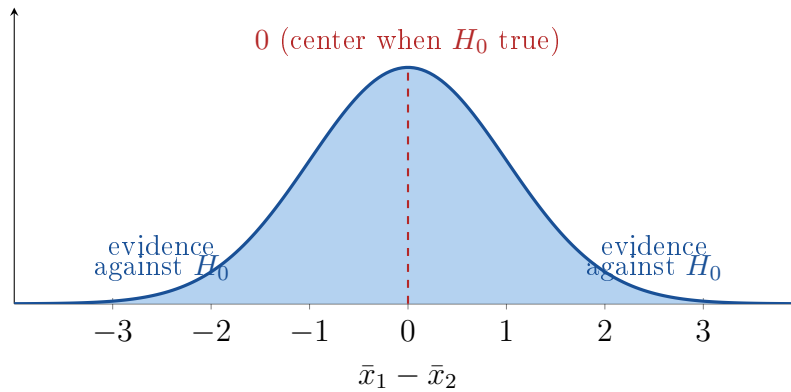
$$\text{SE} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

Visual Insight. Think of it this way: $\bar{x}_1$ is a random variable that varies from sample to sample. So is $\bar{x}_2$. Their difference $\bar{x}_1 - \bar{x}_2$ is also a random variable — and the CLT guarantees it is approximately normal.

When $\mu_1 = \mu_2$ (the null hypothesis), the difference $\mu_1 - \mu_2 = 0$, so the sampling distribution of $\bar{x}_1 - \bar{x}_2$ is centered at zero. The standard error tells us how wide that

distribution is.

Sampling distribution of $\bar{x}_1 - \bar{x}_2$ when $H_0 : \mu_1 = \mu_2$ is true



Observed differences near zero: consistent with H_0 . Differences far from zero (in either tail): evidence against H_0 .

Exercises 7.2

- 7.2.1. Group 1 has $n_1 = 40$, $\sigma_1 = 8$. Group 2 has $n_2 = 50$, $\sigma_2 = 6$. Compute the standard error of $\bar{x}_1 - \bar{x}_2$.
- 7.2.2. If $\mu_1 = \mu_2$, what value do we expect $\bar{x}_1 - \bar{x}_2$ to be close to? What does it mean if the observed difference is very far from this value?
- 7.2.3. Why does the standard error of the difference involve adding the two variances σ_1^2/n_1 and σ_2^2/n_2 , rather than averaging them?

7.3 Two-Sample z -Test for the Difference in Means

Key Formula. Two-sample z -test for $\mu_1 - \mu_2$ (when σ_1, σ_2 known; $n_1, n_2 \geq 30$):

Hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad (\text{equivalently, } \mu_1 - \mu_2 = 0)$$

$$H_a : \mu_1 \neq \mu_2 \quad \text{or} \quad \mu_1 > \mu_2 \quad \text{or} \quad \mu_1 < \mu_2.$$

Test statistic:

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

p-value and conclusion: exactly as in Chapter 6.

The structure is the same as the one-sample z -score: observed minus expected, divided by standard error. Here “observed” is $\bar{x}_1 - \bar{x}_2$, “expected under H_0 ” is 0, and “standard error” is the combined SE.

Example 7.3.1. (*Two-Sample Test for Means: Teaching Methods*) A school tests two teaching methods. Method 1 is used with $n_1 = 35$ students ($\bar{x}_1 = 81.4$, $\sigma_1 = 9.2$). Method 2 is used with $n_2 = 40$ students ($\bar{x}_2 = 76.8$, $\sigma_2 = 11.0$). Test whether the methods produce different mean scores at $\alpha = 0.05$.

Step 1. $H_0 : \mu_1 = \mu_2$. $H_a : \mu_1 \neq \mu_2$ (two-tailed). Both $n_1, n_2 \geq 30$. ✓

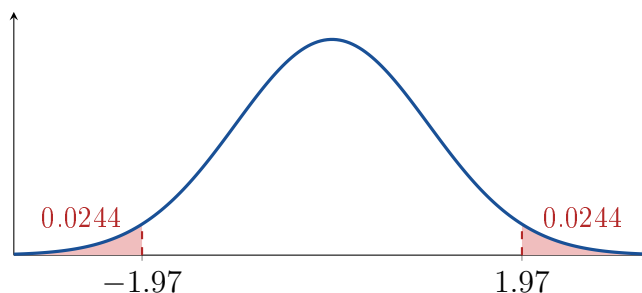
Step 2.

$$SE = \sqrt{\frac{9.2^2}{35} + \frac{11.0^2}{40}} = \sqrt{\frac{84.64}{35} + \frac{121}{40}} = \sqrt{2.418 + 3.025} = \sqrt{5.443} \approx 2.333.$$

$$z = \frac{81.4 - 76.8}{2.333} = \frac{4.6}{2.333} \approx 1.97.$$

Step 3. Two-tailed: $p = 2(1 - P(Z \leq 1.97)) = 2(1 - 0.9756) = 2(0.0244) = 0.0488$.

Two-tailed test: p -value = 0.0488



Step 4. $p = 0.0488 < 0.05 = \alpha$. **Reject H_0 .**

Conclusion: There is sufficient evidence at the 5% level to conclude that the two teaching methods produce different mean scores. Method 1 appears to produce higher scores, though the margin is narrow (p is just below 0.05).

Example 7.3.2. (*One-Sided Two-Sample Test*) A nutritionist compares sodium content in two brands of frozen meals. Brand A: $n_1 = 45$, $\bar{x}_1 = 890$ mg, $\sigma_1 = 120$ mg. Brand B: $n_2 = 38$, $\bar{x}_2 = 840$ mg, $\sigma_2 = 100$ mg. Test whether Brand A has higher mean sodium at $\alpha = 0.01$.

Step 1. $H_0 : \mu_1 = \mu_2$. $H_a : \mu_1 > \mu_2$ (right-tailed).

Step 2.

$$SE = \sqrt{\frac{120^2}{45} + \frac{100^2}{38}} = \sqrt{320 + 263.16} = \sqrt{583.16} \approx 24.15.$$

$$z = \frac{890 - 840}{24.15} = \frac{50}{24.15} \approx 2.07.$$

Step 3. Right-tailed: $p = 1 - P(Z \leq 2.07) = 1 - 0.9808 = 0.0192$.

Step 4. $p = 0.0192 > 0.01 = \alpha$. **Fail to reject H_0 .**

Conclusion: At the 1% significance level, there is insufficient evidence to conclude Brand A has higher mean sodium than Brand B. (Note: the same test at $\alpha = 0.05$ *would* reject H_0 . The significance level matters.)

Exercises 7.3

7.3.1. Two fertilizers are tested on crop yield (bushels per acre). Fertilizer A: $n_1 = 30$, $\bar{x}_1 = 42.1$, $\sigma_1 = 5.8$. Fertilizer B: $n_2 = 35$, $\bar{x}_2 = 39.4$, $\sigma_2 = 6.2$. Test $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 > \mu_2$ at $\alpha = 0.05$.

7.3.2. Using the same data, test at $\alpha = 0.01$. Does the conclusion change? What does this illustrate about the choice of α ?

7.3.3. For a two-tailed test, explain why we write $H_0 : \mu_1 = \mu_2$ rather than $H_0 : \mu_1 - \mu_2 = 0$. Are these equivalent?

7.4 Confidence Interval for $\mu_1 - \mu_2$

Just as in Chapter 5, every two-sample test has a corresponding confidence interval.

Key Formula. A confidence interval for $\mu_1 - \mu_2$:

$$(\bar{x}_1 - \bar{x}_2) \pm z^* \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}.$$

If the interval contains 0, the data are consistent with $\mu_1 = \mu_2$ (fail to reject H_0 at the corresponding level). If the interval does not contain 0, we reject H_0 .

Example 7.4.1. (*CI for the Difference in Means*) Construct a 95% CI for the difference in mean scores between the two teaching methods from Example 7.3.1.

$$(\bar{x}_1 - \bar{x}_2) \pm z^* \cdot \text{SE} = 4.6 \pm 1.96 \times 2.333 = 4.6 \pm 4.57 = (0.03, 9.17).$$

The interval (0.03, 9.17) does not contain 0, confirming that we reject H_0 at $\alpha = 0.05$. We are 95% confident the true difference in mean scores is between 0.03 and 9.17 points in favor of Method 1. The interval is wide, suggesting the true difference could be practically very small or moderately large.

Exercises 7.4

- 7.4.1. Construct a 90% CI for $\mu_1 - \mu_2$ using the fertilizer data from Exercise 7.3.1. Does the interval contain 0? What does this tell you?
- 7.4.2. A 95% CI for $\mu_1 - \mu_2$ is $(-3.2, 8.6)$. (a) Does it contain 0? (b) What is your conclusion about $H_0 : \mu_1 = \mu_2$ at $\alpha = 0.05$? (c) Which group has the larger sample mean? How can you tell?

7.5 Two-Sample z -Test for the Difference in Proportions

We can compare proportions from two groups in exactly the same spirit.

Key Formula. Two-sample z -test for $p_1 - p_2$:

Hypotheses: $H_0 : p_1 = p_2$ vs. $H_a : p_1 \neq p_2$ (or $<$ or $>$).

Pooled proportion (used in the standard error under H_0):

$$\hat{p}_{\text{pool}} = \frac{x_1 + x_2}{n_1 + n_2}$$

where x_1 and x_2 are the counts of successes in each group.

Test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_{\text{pool}}(1 - \hat{p}_{\text{pool}}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}.$$

Conditions: all four counts $n_1\hat{p}_1$, $n_1(1 - \hat{p}_1)$, $n_2\hat{p}_2$, $n_2(1 - \hat{p}_2) \geq 10$.

7.5.1 Why Pool?

Under H_0 , we assume $p_1 = p_2$ — a common unknown proportion p . Our best estimate of that common p is $\hat{p}_{\text{pool}}$, which combines both samples. Using the pooled proportion in the standard error is more efficient than using each sample's proportion separately.

This parallels the one-sample proportion test in Chapter 6, where we used p_0 (the null value) rather than $\hat{p}$ in the standard error.

Example 7.5.1. (*Two-Sample Proportion Test*) A medical study tests whether a new drug reduces the rate of a side effect. In the treatment group, 18 of 120 patients experienced the side effect ($\hat{p}_1 = 0.150$). In the control group, 31 of 110 patients experienced it ($\hat{p}_2 = 0.282$). Test at $\alpha = 0.05$ whether the drug reduces the side-effect rate.

Step 1. $H_0 : p_1 = p_2$. $H_a : p_1 < p_2$ (left-tailed; the drug is expected to *reduce* the rate, so Group 1 rate should be lower).

Check: $120(0.15) = 18 \geq 10$, $120(0.85) = 102 \geq 10$, $110(0.282) = 31 \geq 10$, $110(0.718) = 79 \geq 10$. ✓

Step 2.

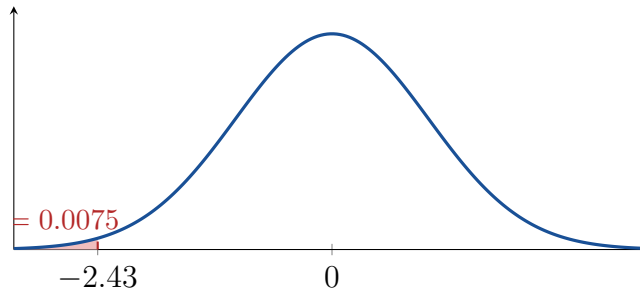
$$\hat{p}_{\text{pool}} = \frac{18 + 31}{120 + 110} = \frac{49}{230} \approx 0.2130.$$

$$\text{SE} = \sqrt{0.2130(0.7870) \left(\frac{1}{120} + \frac{1}{110} \right)} = \sqrt{0.1676 \times 0.01758} = \sqrt{0.002947} \approx 0.05429.$$

$$z = \frac{0.150 - 0.282}{0.05429} = \frac{-0.132}{0.05429} \approx -2.43.$$

Step 3. Left-tailed: $p = P(Z \leq -2.43) = 0.0075$.

Left-tailed test: $p\text{-value} = 0.0075$



Step 4. $p = 0.0075 < 0.05 = \alpha$. **Reject H_0 .**

Conclusion: There is strong evidence at the 5% level that the drug reduces the side-effect rate. The treatment group's rate (15.0%) is significantly lower than the control group's rate (28.2%).

Exercises 7.5

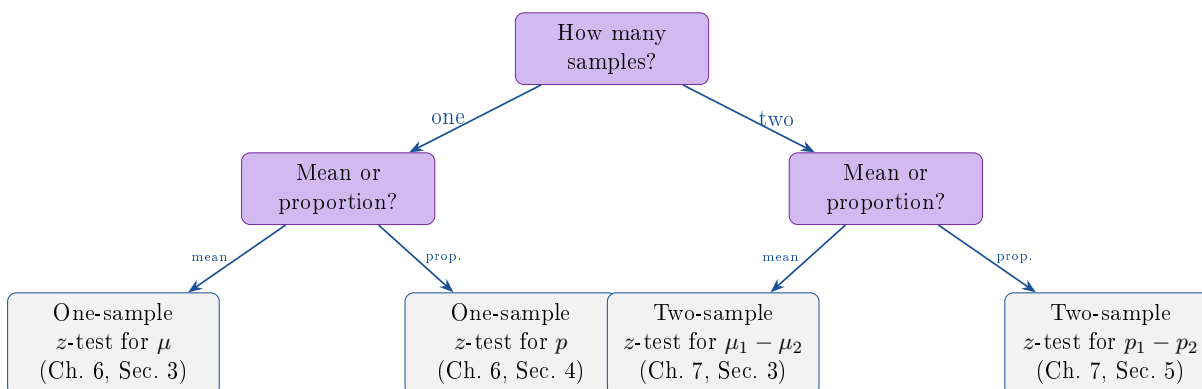
7.5.1. In an election poll, 210 of 400 men support Candidate A, and 195 of 350 women support Candidate A. Test whether support differs by gender ($H_a : p_1 \neq p_2$) at $\alpha = 0.05$.

7.5.2. Why do we pool the proportions under H_0 instead of using each group's $\hat{p}$ separately in the standard error?

7.5.3. A study finds $\hat{p}_1 = 0.45$ ($n_1 = 200$) and $\hat{p}_2 = 0.38$ ($n_2 = 180$). Compute $\hat{p}_{\text{pool}}$ and the test statistic z for a two-tailed test.

7.6 Choosing the Right Test: A Decision Map

With Chapters 6 and 7 complete, we now have four z -tests. Choosing the correct one depends on two questions: (1) one sample or two? (2) mean or proportion?



All four tests share the same structure:

$$z = \frac{\text{observed statistic} - \text{null value}}{\text{standard error}}.$$

The numerator changes (one difference vs. two-sample difference), the denominator changes (one SE formula vs. another), but the logic — compare to the z -table, compare to α — never changes.

Chapter 7 Review

1. A hospital compares recovery times (days) after two surgical procedures. Procedure A: $n_1 = 55$, $\bar{x}_1 = 7.2$ days, $\sigma_1 = 2.1$. Procedure B: $n_2 = 48$, $\bar{x}_2 = 8.0$ days, $\sigma_2 = 2.4$. (a) Test $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 < \mu_2$ at $\alpha = 0.05$. (b) Construct a 95% CI for $\mu_1 - \mu_2$. (c) Confirm that your test result and CI are consistent. (d) Interpret the CI: does it suggest the difference is practically significant, or just statistically significant?
2. A tech company surveys employees in two offices about job satisfaction. Office X: 68 of 100 employees satisfied ($\hat{p}_1 = 0.68$). Office Y: 54 of 90 employees satisfied ($\hat{p}_2 = 0.60$). Test $H_0 : p_1 = p_2$ vs. $H_a : p_1 \neq p_2$ at $\alpha = 0.05$. Follow all four steps.
3. Each of the following study designs produces either paired or independent data. Classify each and explain: (a) SAT scores before and after a prep course, for the same 40 students. (b) Average income of residents in two different zip codes. (c) Cholesterol levels of 50 patients taking Drug A and 60 patients taking Drug B (different patients in each group). (d) Resting heart rate of 30 athletes measured at the start and end of a season.

4. A two-sample test yields $p = 0.03$. A reporter writes: “There is a 97% chance the two groups are genuinely different.” Correct this statement.
5. *Big picture*: In two or three sentences, explain what all four z -tests (one-sample mean, one-sample proportion, two-sample mean, two-sample proportion) have in common structurally, and what changes between them.